

MOLECULAR DYNAMICS SIMULATIONS OF DAMAGE CASCADES CREATION IN OXIDE-PARTICLE-EMBEDDED Fe

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ABSTRACT

Oxide-dispersion-strengthened(ODS)alloys have been identified as one promising candidate alloy type for high temperature reactor applications. Understanding irradiation stability of ODS alloys relies on atomic scale modeling such as molecular dynamics simulations. In this study, yttrium and oxygen charges in Y_2O_3 oxide particles, which are embedded in pure Fe matrix, are optimized to achieve stabilities observed in experiments. Deviation from the optimized charges causes self-explosion and instability of oxide particles. Molecular dynamics simulations further show that under such optimized charge conditions, damage cascade creation and defect developments can be appropriately modeled.

INTRODUCTION

Oxide-dispersion-strengthened (ODS) steels have superior high temperature creep resistance which makes them attractive for advanced reactors [1-5]. Not limited to mechanical property enhancement, oxide particles also influence defect clustering process through possible defect sink properties at oxide-matrix interfaces, hence

leading to possible defect removal capability towards better radiation tolerance. Previous studies have shown that oxide particles are not stable under ion irradiation [2,4]. Oxide particles shows complicated temperature dependence in which the particle sizes shrink under ion irradiation and quickly reach an equilibrium size. The higher the irradiation temperature, the larger the equilibrium sizes [2]. The size dependence on various irradiation conditions depends on the dynamic balances between particle dissolution through damage cascade creation and particle recovery due to diffusions of recoiled Y/O atoms back into particles. In order to understand atomic scale details, molecular dynamics simulations become a necessity [6]. However, the interatomic potentials developed from bulk studies do not consider the complexity of charge issues caused by nanometer sizes and oxide-metal interfaces. The present study aims to provide optimized charges of Y/O atoms in order to accurately describe particles with or without irradiation.

MODELING PROCEDURE

The present modeling uses LAMMPS (Large-scale Atomic Molecular Massively